

DRONE TECHNOLOGY COURSE CONTENT

Duration- 6-8 Weeks

OBJECTIVE:

To provide fundamental and practical knowledge in drone technology, including aerodynamics, components, design, programming, operation, legal regulations, and real-world applications.

WEEK - 01

Introduction to Drone Technology

- History and evolution of drones
- Classification of UAVs (Nano, Micro, Mini, MALE, HALE)
- Fixed-wing vs Rotary-wing drones
- Applications: Agriculture, Surveillance, Delivery, Mapping, Defense
- DGCA/UAS rules and drone categories in India

WEEK - 02

Drone Components & Assembly

- Airframe types and materials
- Brushless DC motors, ESCs, propellers
- Flight controllers: APM, Pixhawk, KK2
- GPS modules, battery types (LiPo), power distribution
- Hands-on: Assembling a quadcopter frame

WEEK - 03

Flight Dynamics & Aerodynamics

- Principles of flight: Lift, Drag, Thrust, Weight
- 6 Degrees of Freedom (DOF)
- Stability, control surfaces, center of gravity
- PID control and tuning
- Hands-on: Testing flight behavior and motor alignment

WEEK - 04

Flight Controller Programming & Sensors

- Understanding Flight Controller firmware (ArduPilot, Betaflight)
- Role of sensors: Gyroscope, Accelerometer, Magnetometer, Barometer
- Mission Planner/ QGroundControl software
- Telemetry, radio transmitters, and receiver calibration
- Hands-on: Configuring flight controller and first test flight

WEEK - 05

Navigation, GPS & Autonomous Flight

- GPS integration and waypoint navigation
- Geo-fencing and return-to-home (RTH) functionality
- Autonomous flight missions using Mission Planner
- Fail-safe modes and emergency protocols
- Practice: Create an autonomous mapping mission

WEEK - 06

Drone Imaging & Payloads

- Types of drone cameras: RGB, Thermal, Multispectral
- Gimbals and stabilization
- Basics of aerial photography and videography
- Real-time data transmission (FPV systems)
- Use in photogrammetry and GIS mapping

OPTIONAL

WEEK - 07

Drone Communication & AI Integration

- RF communication protocols
- Drone swarming basics
- Computer Vision: Object tracking using OpenCV
- Integration with AI models (obstacle detection, route optimization)

WEEK - 08

Project Work & Certification

- Capstone Project: Choose from Agri-drone, Surveillance Drone, Delivery UAV, or Mapping Drone
- Complete assembly, programming, and test flight
- Project report submission
- Evaluation & certification

Technologies & Tools Used

- Hardware: Pixhawk/APM, LiPo battery, 1000KV motors, GPS, ESC, RC TX/RX, telemetry module
- Software: Mission Planner, Q Ground Control, Beta flight, MATLAB/Simulink (optional), OpenCV (optional)
- Simulation (optional): Drone Sim, Gazebo, SITL

Course Deliverables

- Drone design workbook
- Configuration and calibration guide
- Component datasheets and reference manuals
- Mini-project documentation
- Course completion certificate
- Software: Mission Planner, Q Ground Control, Beta flight, MATLAB/Simulink (optional), OpenCV (optional)
- Simulation (optional): Drone Sim, Gazebo, SITL